

Technology #160

The new specific engineering intervention is the integration of a real-time edge computing platform that processes data from various sources and feeds it into the adaptive ensemble model, which then adjusts the EV's powertrain configuration in real-time.

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-09-08

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

The new specific engineering intervention is the integration of a real-time edge computing platform that processes data from various sources and feeds it into the adaptive ensemble model, which then adjusts the EV's powertrain configuration in real-time.

WHY DOES IT MATTER?

Documented problem 'underperformance' concerning electric vehicle. User-submitted engineering problem (session input): "Create an electrical vehicle faster than Tesla Model Y and more efficient than BYD.". ROUTING: nhtsa_complaints: routing state ARCHIVED_ROUTING — measured: NOT_QUERIED_GRAMMAR on every non-vehicle problem (70 recorded grammar mismatches, zero production-run uses in the current domain set — R399 audit W3); reinstatement: the production domain set includes vehicle problems with extracted make|model|year parameters (flip to ACTIVE + worklog entry; connector implementation archived in place). ROUTING: nhtsa_recalls: routing state ARCHIVED_ROUTING — measured: NOT_QUERIED_GRAMMAR on every non-vehicle problem (70 recorded grammar mismatches, zero production-run uses in the current domain set — R399 audit W3); reinstatement: the production domain set includes vehicle problems with extracted make|model|year parameters (flip to ACTIVE + worklog entry; connector implementation archived in place). [nhtsa_complaints] retrieval status NOT_QUERIED_ARCHIVED_ROUTING (engine routing decision (R399 W3): ARCHIVED_ROUTING — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). [nhtsa_recalls] retrieval status NOT_QUERIED_ARCHIVED_ROUTING (engine routing decision (R399 W3): ARCHIVED_ROUTING — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). Problem-existence basis: NO SOURCE RETURNED RECORDS — problem existence UNVERIFIED by live evidence (Art. XX); treat as hypothesis.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in ml_data. 7-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 7-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: The cheapest concrete test that could kill this architecture is to compare its performance to a baseline EV system without real-time adap.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: held-out + temporal validation splits. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: priors and likelihoods are MODEL_DERIVED until validated on deployment data; empirical risk on a non-representative sample says nothing about deployment risk (distribution shift); utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured; ML-002 (Bayes' theorem (posterior inference)) is applicable only under: probability model correctly specified; priors explicit.; ML-003 (Empirical risk) is applicable only under: samples independent and representative of the deployment distribution.; ML-001 (Expected utility of a decision rule) is applicable only under: utility table fully specified; probabilities calibrated.; Model input variables requiring sourced values before any numeric result: $P(x|y)$ (-), $P(y)$ (-), $P(x)$ (-), n (-), loss (-), $f(x_i)$ (-), y_i (-), $P(y|x)$ (-), U (-) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: $P(y|x)$ (-), R_{emp} (-), EU (-) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): training data volume and provenance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; label quality rate $[y_i, -]$ value_status=UNKNOWN; target operating point (precision/recall) [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; calibration error budget [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; inference latency ceiling [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; distribution-shift monitoring thresholds [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: The new causal mechanism is a real-time adaptive ensemble machine-learning model that incorporates diverse data sources, including sensor data, weather forecasts, traffic information, and reinforcement learning algorithms for adaptive battery management.

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'underperformance' concerning electric vehicle. User-submitted engineering problem (session input): "Create an electrical vehicle faster than Tesla Model Y and more efficient than BYD.". ROUTING: nhtsa_complaints: routing state ARCHIVED_ROUTING — measured: NOT_QUERIED_GRAMMAR on every non-vehicle problem (70 recorded grammar mismatches, zero production-run uses in the current domain set — R399 audit W3); reinstatement: the production domain set includes vehicle problems with extracted make|model|year parameters (flip to ACTIVE + worklog entry; connector implementation archived in place). ROUTING: nhtsa_recalls: routing state ARCHIVED_ROUTING — measured: NOT_QUERIED_GRAMMAR on every non-vehicle problem (70 recorded grammar mismatches, zero production-run uses in the current domain set — R399 audit W3); reinstatement: the production domain set includes vehicle problems with extracted make|model|year parameters (flip to ACTIVE + worklog entry; connector implementation archived in place). [nhtsa_complaints] retrieval status NOT_QUERIED_ARCHIVED_ROUTING (engine routing decision (R399 W3): ARCHIVED_ROUTING — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). [nhtsa_recalls] retrieval status NOT_QUERIED_ARCHIVED_ROUTING (engine routing decision (R399 W3): ARCHIVED_ROUTING — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). Problem-existence basis: NO SOURCE RETURNED RECORDS — problem existence UNVERIFIED by live evidence (Art. XX); treat as hypothesis.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: ml_data

Disciplines: data engineering, machine learning, software engineering, statistics

Architecture: PROPOSED architecture (no physical system exists): data ingestion and labeling pipeline; feature / representation layer; model + uncertainty quantification; deployment target + monitoring loop — realizing: The new specific engineering intervention is the integration of a real-time edge computing platform that processes data from various sources and feeds it into the adaptive ensemble model, which then adjusts the EV's powertrain configuration in real-time.

Subsystems:

?: data ingestion and labeling pipeline - ?

?: feature / representation layer - ?

?: model + uncertainty quantification - ?

?: deployment target + monitoring loop - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: The new specific engineering intervention is the integration of a real-time edge computing platform that processes data from various sources and feeds it into the adaptive ensemble model, which then adjusts the EV's powertrain configuration in real-time.

Key Physics: Data architecture / machine learning governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Data architecture / machine learning governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

ML-002: Bayes' theorem (posterior inference). Expression: $P(y|x) = P(x|y) * P(y) / P(x)$. Applicability to this invention: APPLICABLE; holds under: probability model correctly specified; priors explicit. Assumptions: priors and likelihoods are MODEL_DERIVED until validated on deployment data. Engaged variables: $P(x|y)$.

ML-003: Empirical risk. Expression: $R_{emp} = \text{sum}(\text{loss}(f(x_i), y_i)) / n$. Applicability to this invention: APPLICABLE; holds under: samples independent and representative of the deployment distribution. Assumptions: empirical risk on a non-representative sample says nothing about deployment risk (distribution shift). Engaged variables: $n, f(x_i)$.

ML-001: Expected utility of a decision rule. Expression: $EU = \text{sum}_y P(y | x) * U(a(x), y)$. Applicability to this invention: APPLICABLE; holds under: utility table fully specified; probabilities calibrated. Assumptions: utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured. Engaged variables: EU.

Assumptions:

- priors and likelihoods are MODEL_DERIVED until validated on deployment data
- empirical risk on a non-representative sample says nothing about deployment risk (distribution shift)
- utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured
- ML-002 (Bayes' theorem (posterior inference)) is applicable only under: probability model correctly specified; priors explicit.
- ML-003 (Empirical risk) is applicable only under: samples independent and representative of the deployment distribution.
- ML-001 (Expected utility of a decision rule) is applicable only under: utility table fully specified; probabilities calibrated.
- Model input variables requiring sourced values before any numeric result: $P(x|y)$ (-), $P(y)$ (-), $P(x)$ (-), n (-), loss (-), $f(x_i)$ (-), y_i (-), $P(y|x)$ (-), U (-) — all UNKNOWN (no sourced value exists; Art. VIII).

- Model output variables produced by the governing model: $P(y|x)$ (-), R_{emp} (-), EU (-) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): training data volume and provenance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; label quality rate $[y_i, -]$ value_status=UNKNOWN; target operating point (precision/recall) [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; calibration error budget [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; inference latency ceiling [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; distribution-shift monitoring thresholds [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'underperformance', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'distribution shift degrading accuracy post deployment', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'label noise propagating to biased predictions', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'overfitting to source-site artifacts', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'calibration drift invalidating decision thresholds', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'feedback-loop data contamination', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	exceed Tesla Model Y speed and BYD efficiency	SOURCE_FACT
DI-002	Mechanism driver (proposed)	The new causal mechanism is a real-time adaptive ensemble machine-learning model that incorporates diverse data sources	MODELED
DI-003	Expected effect target	The new architecture is expected to improve the EV's efficiency by 15% and model speed by 10% compared to the reference	MODELED
DI-004	Domain design input pattern: decision-the applicability of support tokens [decision]	ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: available applicability of data established (candidate pattern; confirm again)	ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: latency applicability of established (candidate pattern; confirm again)	ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance candidates standards for OEMs (Road Ministry, Highway, PKB, OEV, UN, etc.)	ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-008	Custodied observation: Adaptive state health monitoring	https://arxiv.org/abs/2005.07141	144589c6ae239ee8
DI-009	Custodied observation: Path informed adaptive deployment	https://arxiv.org/abs/2005.07141	0db67428db4e5d
DI-010	Custodied observation: An integrated deep learning framework	https://arxiv.org/abs/2005.07141	ca8cea1096e5a769
DI-011	Custodied observation: Hybrid multi-scale CNN	https://arxiv.org/abs/2005.07141	80ea5d0ac4b0f7d5
DI-012	Custodied observation: Preventing Blackouts caused by Frequency	https://arxiv.org/abs/2005.07141	24caf30dad7f393c9

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	data ingestion and labeling pipeline realizing The new specific engineering intervention	ENGINEERING	details of the integration of real-time edge computing platform that processes
DO-002	feature / representation layer realizing The new specific engineering intervention	ENGINEERING	details of the integration of real-time edge computing platform that processes
DO-003	model + uncertainty quantification realizing The new specific engineering intervention	ENGINEERING	details of the integration of real-time edge computing platform that processes
DO-004	deployment target + monitoring loop realizing The new specific engineering intervention	ENGINEERING	details of the integration of real-time edge computing platform that processes
DO-005	data ingestion and labeling pipeline interfaces with feature / representation layer	ENGINEERING	signature, granularity (name of files, the new specific engineering intervention)
DO-006	feature / representation layer interfaces with model + uncertainty quantification	ENGINEERING	signature, granularity (name of files, the new specific engineering intervention)
DO-007	model + uncertainty quantification interfaces with deployment target + monitoring loop	ENGINEERING	signature, granularity (name of files, the new specific engineering intervention)
DO-008	Device boundaries: patient/fluid inlet and outlet terminal control/data processing	ENGINEERING	signature, granularity (name of files, the new specific engineering intervention)
DO-009	Flow/force/signal path geometry realizing the mechanism	ENGINEERING	signature, granularity (name of files, the new specific engineering intervention)
DO-010	Operating range for training data volume and provenance	ENGINEERING	signature, granularity (name of files, the new specific engineering intervention)
DO-011	Operating range for label quality rate within which the mechanism holds its stated behavior	ENGINEERING	signature, granularity (name of files, the new specific engineering intervention)
DO-012	Operating range for target operating point (precision/recall)	ENGINEERING	signature, granularity (name of files, the new specific engineering intervention)
DO-013	Operating range for calibration error budget within which the mechanism holds its stated behavior	ENGINEERING	signature, granularity (name of files, the new specific engineering intervention)
DO-014	Operating range for inference latency ceiling within which the mechanism holds its stated behavior	ENGINEERING	signature, granularity (name of files, the new specific engineering intervention)
DO-015	Operating range for distribution-shift monitoring thresholds	ENGINEERING	signature, granularity (name of files, the new specific engineering intervention)
DO-016	Sensing -> decision -> actuation loop maintaining the proposed operating point	ENGINEERING	signature, granularity (name of files, the new specific engineering intervention)
DO-017	Information pipeline: ingestion -> feature/representation -> UQ -> decision	ENGINEERING	signature, granularity (name of files, the new specific engineering intervention)
DO-018	Fixture enabling: held-out + temporal validation splits	ENGINEERING	signature, granularity (name of files, the new specific engineering intervention)
DO-019	Fixture enabling: calibration and decision-threshold analysis	ENGINEERING	signature, granularity (name of files, the new specific engineering intervention)
DO-020	Fixture enabling: failure-mode stress tests (distribution shift)	ENGINEERING	signature, granularity (name of files, the new specific engineering intervention)

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: held-out + temporal split evaluation	UNKNOWN	Unknown

Unknown: pit named, build specification, instrumentation selection; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
training data volume and distribution	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
label quality rate	UNKNOWN (no sourced value)	-	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
target operating point (precision)	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
calibration error budget	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
inference latency ceiling	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
distribution-shift monitoring thresholds	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
underperformance	Fails by underperformance: the device fails to hit a target accuracy (measured on test set) due to insufficient training data, poor model architecture, or hardware constraints.	Design for failure: ensure sufficient training data and reserve hardware resources for worst-case deployment.	UNKNOWN (no sourced value)	UNKNOWN (no sourced value)
distribution shift degradation	Fails by distribution shift: the model's performance degrades when deployed on data from a different source than the training data.	Design for failure: ensure the model is robust to distribution shifts by using techniques like domain adaptation or ensemble methods.	UNKNOWN (no sourced value)	UNKNOWN (no sourced value)
label noise propagating	Fails by label noise: the model's performance degrades due to noisy labels in the training data, which propagate through the model.	Design for failure: ensure the model is robust to label noise by using techniques like loss functions that are less sensitive to outliers.	UNKNOWN (no sourced value)	UNKNOWN (no sourced value)
overfitting to source	Fails by overfitting to source: the model performs well on the training data but poorly on the test data, indicating it has learned the noise in the training data.	Design for failure: ensure the model is robust to overfitting by using techniques like regularization or cross-validation.	UNKNOWN (no sourced value)	UNKNOWN (no sourced value)
calibration drift invalidation	Fails by calibration drift: the model's predicted probabilities become increasingly inaccurate over time, leading to poor decision-making.	Design for failure: ensure the model is robust to calibration drift by using techniques like temperature scaling or dynamic thresholding.	UNKNOWN (no sourced value)	UNKNOWN (no sourced value)
feedback-loop data corruption	Fails by feedback-loop data corruption: the model's performance degrades due to corrupted data being fed back into the training loop.	Design for failure: ensure the model is robust to data corruption by using techniques like data validation and anomaly detection.	UNKNOWN (no sourced value)	UNKNOWN (no sourced value)

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
underperformance	Fails by underperformance: the device	design controls	NOT ESTABLISHED (defined after verification)
distribution shift degrading accuracy	Fails by distribution shift degrading accuracy	cross deployment triggers	On post-deployment, we test against the training data (Article X)
label noise propagating to false predictions	Fails by label noise propagating to false predictions	error detection systems	we use the method based on gradient map (Article X)
overfitting to source-site artifacts	Fails by overfitting to source-site artifacts	site-level key data acquisition	validation on the source site is provided (Article X)
calibration drift invalidating decisions	Fails by calibration drift invalidating decisions	calibration thresholds	get feedback by monitoring the ratio of changed data (Article X)
feedback-loop data contamination	Fails by feedback-loop data contamination	external judgement	sample data from other sites to ensure data quality (Article X)

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	The cheapest concrete test that could kill the hypothesis	kill the hypothesis to test	kill the hypothesis to test	kill the hypothesis to test
VF-002	run killer experiment: falsification_test	kill the hypothesis to test	kill the hypothesis to test	kill the hypothesis to test
VF-003	held-out + temporal validation splits. Acceptance criteria	held-out + temporal validation splits	held-out + temporal validation splits	held-out + temporal validation splits
VF-004	held-out + temporal validation splits. Acceptance criteria	held-out + temporal validation splits	held-out + temporal validation splits	held-out + temporal validation splits
VF-005	failure-mode stress tests (distribution shift)	failure-mode stress tests	failure-mode stress tests	failure-mode stress tests

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	acceptance criteria to be fixed at validation	NOT ESTABLISHED (defined after validation)

11. Materials

Component	Candidate Material	Source	Verification Required	Status
NOT_APPLICABLE — data only invention: no physical materials exist to select; hardware dependencies are recorded under design	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
NOT	ESTABLISHED (no domain data)	NOT	ESTABLISHED (no domain data)				

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- MLOps CI for data and model: domain registry manufacturing route - candidate only; no process qualification exists
- versioned artifact registry with rollback: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-002', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-004', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter training data volume and provenance [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter label quality rate [y_i -]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter target operating point (precision/recall) [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter calibration error budget [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter inference latency ceiling [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter distribution-shift monitoring thresholds [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}

- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}
- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-005', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-005'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article implementation of the proposed physical verification process	per method	maintaining the proposed physical verification process	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-02	constraint-compliance envelope article	per method	constraint-compliance envelope article	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	The cheapest concrete	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment: NOT ESTABLISHED	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	held-out + temporal validation	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	held-out + temporal validation	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	failure-mode stress test	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 7-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.